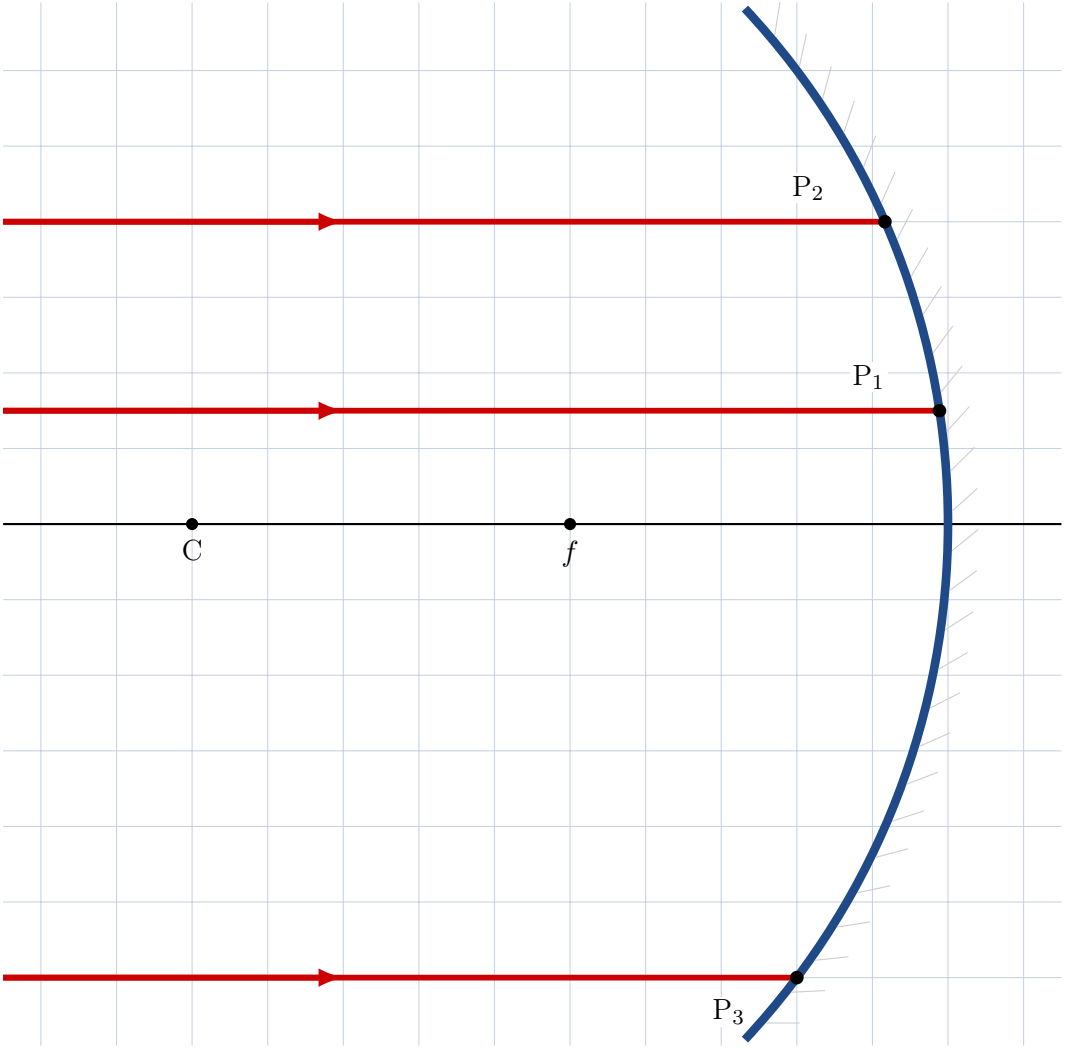
1. Why is $f = R/2$? Below is a concave mirror drawn to scale: a slice of a circle with radius $R = 10$ cm, centered at C. Three rays come in parallel to the axis at different heights and hit the mirror at P_1, P_2 and P_3. One square is 1 cm.



A. The **normal** at P_1 is the line from C through P_1. Draw it. Why is that line perpendicular to the mirror?

B. Use a protractor to measure θ_i between ray 1 and its normal. Then draw the reflected ray with $\theta_r = \theta_i$, and mark where it crosses the axis.

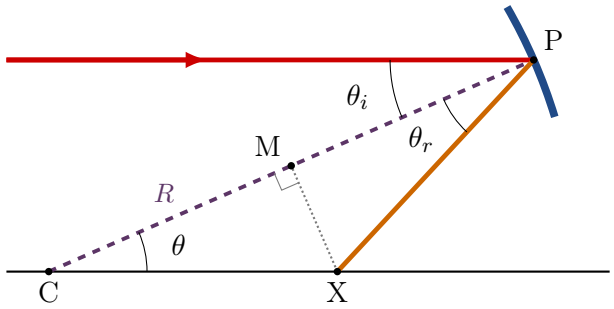
C. Repeat for rays 2 and 3. Fill in the table (distances measured from the mirror).

ray	height	θ_i	crosses the axis at
1	1.5 cm	_____	_____
2	4 cm	_____	_____
3	6 cm	_____	_____

D. Which ray lands closest to $f = R/2 = 5$ cm? Which misses by the most, and does it cross **closer to** or **farther from** the mirror than f?

E. A spherical mirror does **not** bring every parallel ray to one point. The blur this causes is called **spherical aberration**. The mirror equation (and $f = R/2$) only works for **paraxial** rays: rays close to the axis. Use your table to explain why.

F. Now calculate the miss. Take ray 2 ($h = 4$ cm). Call the point where its reflected ray crosses the axis X. The normal CP, the reflected ray PX and the axis CX make a triangle. M is the midpoint of CP.



(i) CP is a radius, so CP $= R = 10$ cm. Use $\sin\theta = h/R$ to find θ. Does it match your protractor?

(ii) The incoming ray is parallel to the axis. Why is the angle at C also θ?

(iii) The angle at P between the normal and the reflected ray is θ_r. Explain why triangle CXP is **isosceles**, so CX = XP.

(iv) Because CX = XP, the line XM cuts the triangle into two right triangles, and CM $= R/2$. Use $\cos\theta$ to find CX.

(v) How far is X from the mirror? How far from f? Compare with your table.

(vi) For a ray very close to the axis, θ is tiny and $\cos\theta \approx 1$. What does CX become? What does that tell you about f?

(vii) Put it together: write one formula for the miss from f that works for **any** θ, in terms of R and θ. (Hint: $\sec\theta = 1/\cos\theta$.)

2. Graph the miss. Open GeoGebra or Desmos and graph your formula from part F (vii) with $R = 10$ cm, using x for the angle in degrees.

Hints

	GeoGebra	**Desmos**
Degrees	write `x°` inside the trig function (type `Alt+O` for °)	wrench (Graph Settings) → **Degrees**
Constant	`R = 10`	`R = 10`
The miss as a graph	`m(x) = R/2 (sec(x°) - 1)`	`m(x) = R/2 (sec(x) - 1)`
Slider for θ (call it t)	`t = Slider(0, 60, 0.1)`	`t = 20`, then click "add slider" and set 0 to 60
A point that rides the curve	`P = (t, m(t))`	`(t, m(t))`

Set the window to $0 \leqslant x \leqslant 60$ and $0 \leqslant y \leqslant 6$.

A. Slide t to the three angles from your table. Do the miss values match what you measured and calculated?

B. Sketch the shape of the graph. Why is it so flat near $\theta = 0$?

C. You want every ray to land within 0.5 cm of f. Slide t to find the biggest θ that works. How far from the axis is that ray ($h = R\sin\theta$)? How tall can the mirror be?

D. Change R to 20 cm. What happens to the miss at the same angle?